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YOLO Algorithm Implementation for Real Time Object Detection and Tracking

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***Abstract*— Nowadays modern society is over flooded with humongous masses of visual data. There exist many image analysis methods to dive into this sea of visual information. The constituents of these images and videos can be analyzed and further processed to recognize the useful information. The detection, identification, and localization of different objects could prove to be of mammoth use and can play a significant part in modern devices and technologies. This paper represents a comparative study of several entity recognizing methods like YOLO, Faster R-CNN and R-CNN over different parameters such as mAP, FPS, etc. This paper also introduces an intelligent system (robot) that is capable of localizing an object and following it in real-time. The required input image is provided by the ESP32 cam module which can be mounted on the robot. Machine Learning Algorithms are used for object detection. Position coordinates received are then used to locate, track and follow the moving object. Furthermore, it is of interest as it can scale down human tasks and help mortals to be aware of minute details about certain objects.**

Keywords*— *Object Detection, Object Tracking, Machine Learning, Deep Learning, Comparative Study, RCNN, Faster RCNN, Computer Vision, Image Processing, YOLO, ESP32.

I. INTRODUCTION

Humans have the capability to see and identify objects present in their surroundings. They also have the intelligence to make decisions and choose whom they want to interact with and how. With the recent increase in the need of automation, there has also been a rise in robotic development technology. Many researches have been carried out in a similar domain that aims at improving automation and integrating it with our daily life. With the rise in the domain of Machine Learning, more and more integration has been happening. Computer Vision based intelligent robots are one such use case.

Computer Vision is a branch that deals with Machine Learning of visual data. The aim is to create an intelligent robot that can take visual data, process it, and then can follow a specific object.

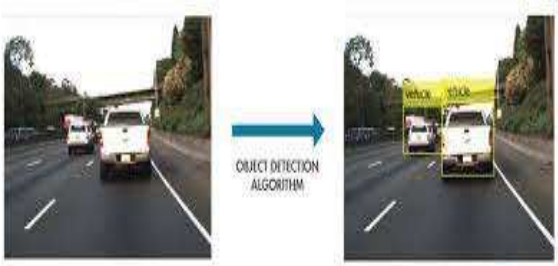

Fig. 1. Object Detection on the image

Machine Learning has been used in the software part of this project. Object detection techniques are used for visual data preprocessing. The YOLO algorithm is applied for object detection in the frame. Instead of using pretrained weights of YOLO, we did custom training of YOLO. Custom Training is performed on images of balls that have been collected

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from open source [20]. Input images are captured through ESP32 cam and sent to a laptop which is connected through Wi-Fi. In the laptop the Machine Learning model detects the location of the object inside the image frame and returns the position coordinates in 2-dimension. Once the position coordinates are received, this information can be used to turn the motor of the robot in a specific direction. This way the robot can follow an object which it detects.

Any deep learning object detection model using sectional anchor boxes can be branched into two stages. Candidate box election is the first and foremost step followed by abstraction and detection of features. However, Faster R-CNN have accomplished independent training. But, due to the presence of innumerous training parameters and complicated neural network structure these models are hard-won to identify in real time. On the other hand YOLO sections the image into S×S sized grids, and then anticipates the bounding box and class for every grid. This eliminates the proposition of candidate regions and thus makes the network comparatively less complicated, which leads to considerable improvement in the detection speed.

This paper presents the novel idea of implementation of Object Detection and Tracking using the YOLO algorithm in real-time. A combination of Robotics and Machine Learning has been used. It has been integrated with a moving robot that can detect and follow objects.

Rest of the paper represents: Section II reviews the related work in the concerned domain. The methodology proposed is explained through section III. Hardware tools and Software technologies used are illustrated in section IV. Next section describes the experimentation and results. Section VI depicts the comparative study of various object detection techniques. Finally, section VII forms the conclusion and future scope of the project.

II. LITERATURE REVIEW

The sufficient amount of literature has been reviewed in the related domains. These are described as follows:

Joseph Redmon et al., presents YOLO as a novel scheme for object detection, as formerly object detection was only seen as classification for performing detection [9]. For the prediction of possibilities of different classes using a single neural network, a structure of the retrogress problem was formed. The whole architecture is very fast as the entire model is trained jointly.

Hasan et al., proposed an object trailing robot which uses image data and takes instructions from a computer to identify and trail a colored object [1]. The robot uses a webcam which is connected to the computer for visual data input, then instructions are generated from the computer after the processing of image data to demonstrate physical movements. This paper is divided into the following three parts: image processing, object tracking algorithm and mechanism of steering the robot.

Xiyang Song et al., integrates Machine Learning Algorithms with robots [2]. The resultant of this is a smart autonomous robot which can navigate successfully in an unknown environment to reach its destination. A BP neural network trained on several samples is used as a navigation system. This robot uses 5 ultrasonic sensors to gather data.

M. S. Sefat et al., presents the idea of making a vision-based robot that can both follow as well as track a certain object based on the users' command [3]. It uses Kalman filter for better position estimation of the object and the Hybrid Automata Model to act according to the situation. This paper further describes how this robot can be further implemented to perform object picking such as picking waste, picking a certain fruit, etc.

Yuxin Jing et al., presents a Remote Control (RC) Car unit that can be used in search and rescue operations in disaster-struck areas [4]. This unit is compact and can easily fit in small spaces between the rubble to search and transfer video and data on the location and condition of the survivors to the rescuers. It has an ultrasonic sensor to detect obstacles and it can be controlled by an android device only.

Katherine Rose Driggs-Campbell et al., summarizes the work that has been done in the areas of vehicle-to-vehicle communication [5]. It talks about various autonomous decision-making algorithms and many different practical scenarios which should be considered. It also explains different control applications and various high-level decision-making algorithms that need to be refined and then implemented to make semi-autonomous and fully autonomous vehicles a reality.

Kishan Kumar et al., presents the idea of using voice commands to enhance the applicability of an object tracking robot towards physically challenged people [7]. The robot maneuvers around as per the voice-command given, while also keeping the track of the desired object. The input voice command signals are being processed in real time on a cloud server. However this robot faces the issue of a small working range of 10-12m due to using bluetooth technology.

Nishchal K. Verma et al., gives the idea of using Compressive tracking on a vision-based object following system [8]. This paper explains how a compressive tracking Algorithm is used to track the object and how using Stereo-vision we can extract 3D data from 2D images using 2 pinhole cameras. This paper concludes that the application of these algorithms in a vision-based system results in increased computational efficiency, accuracy, and robustness.

M. N. Vijayalakshmi et al., evaluates the performance of object detection techniques [10]. Pre-existing template matching, color and shape-based techniques are implemented on visual data and then

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correlated under a variety of scenarios to find which identifier is sturdy under different circumstances.

Rachna Verma is presenting a comprehensive review of different object tracking and object detection technologies [11]. This paper explains various pre-established technologies and proposes a new way to categorize them to group various detection and tracking approaches, which will make future research more streamlined. It also briefly explains various methods used in the categorization of the object detection and tracking approaches.

Galandaru Swalaganata et al., presents a Hybrid method to track objects [13]. To make the object tracking outcome more accurate, the prediction method is affiliated with the Kalman filter. This approach largely improved the overall performance of the tracking system.

Zong-Qiu Zhao et al., introduces new methods for detecting objects [14]. It compares the performance of traditional object detection methods and provides improvement for increasing efficiency and accuracy. It also shows various characteristics of face detection and object detection.

E. A. Oyekanlu et al., discusses the research done in the past as well as the technologies presently being used in developing Mobile Robots and guided Vehicles in the present [15]. This paper discusses various Localization Navigation and Control Algorithms presently being used to develop AVGs and AMRs. It also discusses various areas in which research is needed for the application of 5G networks in various technologies.

Krishnakumar Marapalli et al., introduces an intelligent vehicle that can monitor remote locations for border surveillance and to identify enemies at wars [16]. Different technologies such as wireless communication, IOT, software, and mobile applications are applied. Certain hand gestures are specified for different directions which are used to navigate the constructed robot. This human-computer interaction device is cost efficient, scalable, and efficient to aid the military where conditions are not favorable and risky for humans to work.

N. Murali Krishna et al., proposed YOLOv3 as a formidable example of a single stage object identifier [17]. For object detection, famous Convolutional Neural Networks (CNN) based algorithms exist like RCNN and FRCNN. These algorithms may beat YOLO in terms of accuracy but if speed is taken as a deciding parameter, then YOLO has superiority. It is exceedingly fast as it only takes one pass of the artificial neural network.

Ramasamy et al., discusses various security issues that can be faced by a mobile robot [18]. Since all mobile robots depend on the sensors to gather information about their surroundings, these sensors are vulnerable to different kinds of exploits e.g. ultrasonic sensors are vulnerable to jamming attacks, etc. This paper discusses such threats and also provides recommendations for the successful implementation of a secure mobile robot.

III. PROPOSED METHODOLOGY

The block diagram shown below in Figure 2 describes the complete workflow of the Object Tracking Robot. It also depicts how multiple devices are connected with each other.

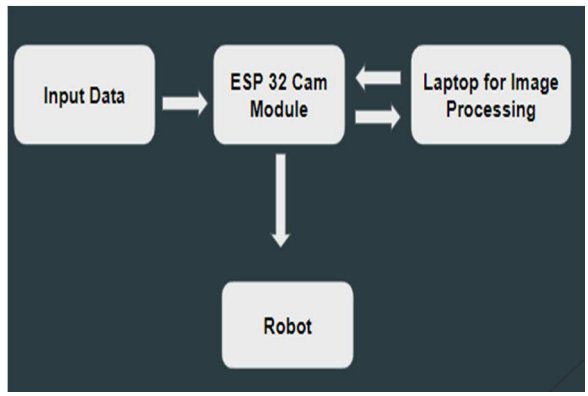

Fig. 2. Block Diagram of the proposed system

The presented Robot consists of ESP32 CAM Module which is used to capture surrounding image frames from a live video and those images are later sent to the computer for further processing using Python and intelligent models for detecting desired objects. After detecting the desired object it will calculate the distance and direction that is required to move in order to reach the object. These instructions are sent back to the ESP32 CAM Module which will then control the DC Motors through the L293D motor driver circuit and move in the required direction to follow the object.

IV. TOOLS AND TECHNOLOGY

A. *Hardware Components used:*

1) *ESP32-CAM:* It is a small size, low power consumption, low-cost development board. It has an ESP32-S processor, a microSD card slot and an OV2640 camera. It supports WiFi video monitoring and WiFi image upload. Here, it is used to take image input for our ML Model.
2) *FTDI Programmer:* It is a converter module, it converts USB to TTL serially. Here the FTDI board is used to upload the code.
3) *L298N Motor Driver Module:* It used to drive DC motors. It can drive upto 2A of current.

B. *Software Technology used:*

1) *Python:* It is the most widely used programming language in the field of Artificial Intelligence and Machine Learning.
2) *OpenCV:* The most common Computer Vision and Image Processing is used. All the preprocessing of images in a particular frame is done with the help of functions present in the

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OpenCV library. The text displayed on the screen is also done with the help of OpenCV.

3) *Google Colab:* It supplied free GPU and TPU which is very necessary for training any Machine Learning Algorithm. This platform is really helpful for those people who do not have proper resources for training such heavy models.We have also used Google Colab for the custom training of the YOLO algorithm.
4) *Arduino IDE:* It is available for every operating system like Linux, Windows, MacOs.

C. Algorithms Used:

1) *Object Detection:* It is a technique in which instances of objects in images or videos are localized [17]. In simpler terms, it deals with identifying and locating some of the classes such as person, car, bus, ball, etc. from the image. It basically counts the number of instances of unique objects in a particular image and shows their precise locations, along with labeling.

   Many Algorithms are used to perform object detection on an image and produce meaningful results from it. This paper introduces YOLO as an Object Detection algorithm for Object Tracing Robots.

2) *YOLO:* YOLO is an object detection algorithm. Earlier many algorithms were used by people such as sliding window object detection then switched to RCNN, fast RCNN and faster RCNN [18]. But after the invention of YOLO, it has become the most used algorithm for object detection and the main reason being its speed.It is not only fast but also accurate. During real time, detection speed plays a crucial role,that is why YOLO is the best option.

How does YOLO work?

YOLO works on the basic idea of dividing images into smaller squares called cells. Every image frame is divided into a small matix and that section is responsible for detection. Dependent and independent variables are provided to the model for training. Independent variable is the image on which object detection has to be performed and dependent variables consist of the position coordinates. The dependent feature consists of 5 variables, x position, y position, width, height, and confidence score which can be 0 or 1.

Once the YOLO model is trained on these features, weights can be saved which can later be used for detection in real-time.

V. EXPERIMENTS AND RESULTS

The real-time Ball detection model works on YOLO. YOLO performs entity detection considering it as a regression problem and provides the probability of the predicted class. The YOLO algorithm uses CNN or convolutional neural networks for object detection in real-time. Following are the steps to be followed for the training of the model:

- During the training process, the YOLO model was custom trained over three thousand images, and post training the weights were saved and used for object detection [19].
- After training the model with the saved weights, the input images are captured from the ESP32 camera module and are displayed on a local web server created by the ESP32 Wi-Fi module and it is used locally for further processing.
- These profiles are then sent to the Machine learning model for performing entity detection and to get 2-D position coordinates of the object as the output of each frame.
- The Machine Learning model was tested on objects (balls) of different sizes and colors and as a result it successfully detected the balls with a good prediction probability in the video frame and it gave position coordinates of the object as output of each frame.
- The accuracy of the designed model over a given dataset is found out to be 66.77% as shown in figure 8.

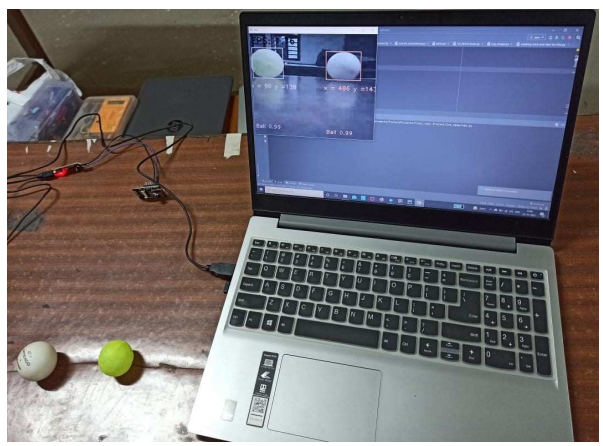

Fig. 3. Demonstration of Experimental set-up

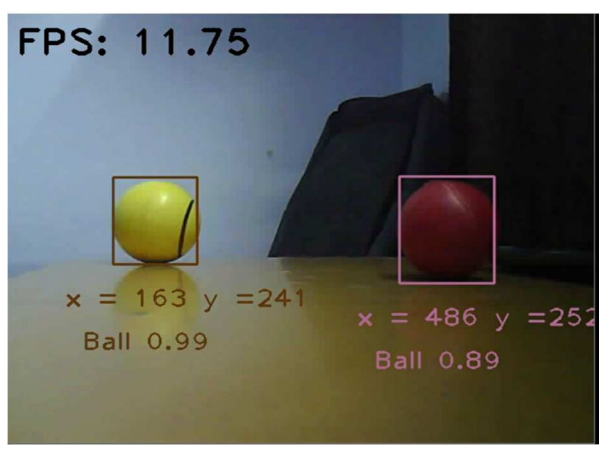

Fig. 4. Object Detection (Balls) by the Model

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In Figure 4 Object detection by the Model is displayed. Bounding Boxes appear when a ball is detected. It shows the x and y position coordinates. The probability of the detected object to be a ball is 0.99 and 0.89 for the Yellow and Red ball respectively. The Video Demonstration of object detection [21] and the working robot following the object [22] is also available.

VI. COMPARISON

A. *RCNN:* Stands for Region with Convolutional Neural Network, it coalesces convolutional neural network features with box sized section schemes. The R-CNN algorithm will initially look for sections in the image frame that may enclose an item, which are referred to as region suggestions. The CNN features will then be calculated from the region proposals. It will then categorize the items based on the extracted features in the last stage [6].

B. *FASTER RCNN:* Faster R-CNN exploits the RPN (Region Proposal Network) that allots complete image convolutional features with the identification network in a more economical demeanor than Fast R-CNN and R-CNN. In faster R-CNN, the region proposal neural network inputs convolutional attributes from the convolutional neural network and computes it to identify a particular object, rather than searching over, for the object in the image again [12].

Comparison of models is usually based on certain parameters. Resources required, time taken and accuracy are one of the most important of those parameters. In case of real time detection Latency also plays a major role. Each term is described in detail below:

- Latency: It is a measure of time taken to process one unit of data for batch size 1. Latency is significant as it directly shows the real time performance of a system. Thus it can be said that less latency is better.
- mAP: It stands for Mean Average Precision. It is a known metric in calculating accuracy in object detection techniques.
- FPS: It is an abbreviation for 'Frames Per Second.' It is the number of consecutive images that can be detected by a model each second.

Table 1 shows the comparison between different object detection techniques on the basis of various parameters.

Table I. Comparison between different object detection techniques

Model	**Latency**	**mAP**	**FPS**	**Real Time**	**Dataset**
R-CNN	Medium	~54	< 1	No	PASCAL-VOC 2010
Faster R-CNN	Medium	~70	5-17	No	PASCAL-VOC 2007
YOLO	Low	67	25-30	Yes	Custom Dataset

VII. CONCLUSION

The paper presents a simple, competent and robust methodology which can be implemented with any object following system to perform object detection and tracking.

This method uses the YOLO Algorithm which can perform both classification and prediction of bounding boxes by only using one neural network. This algorithm can heavily optimize the detection performance. It is much lighter and faster than the standard Machine Learning Algorithms. The algorithms are judged on accuracy, robustness and computing efficiency. In this paper a comparative study is done on various object identifying methodologies like RCNN, Faster RCNN and YOLO. We found that YOLO is much faster and accurate than other object detection techniques.

The intelligent robot designed is capable of detecting and following an object in real-time. The required input image is provided by the ESP32 cam module which can be mounted on the robot. It is of interest as it can scale down human tasks and help us to be aware of minute details about certain objects.

This could also be implemented in human following robots where object detection could be trained for humans instead of any non-living object. In that case, the machine learning algorithm should be provided with a wide variety of datasets containing different human positions in different environments. This technology could be utilized and implemented in a number of different areas, such as security surveillance, military applications, and child monitoring, etc.

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